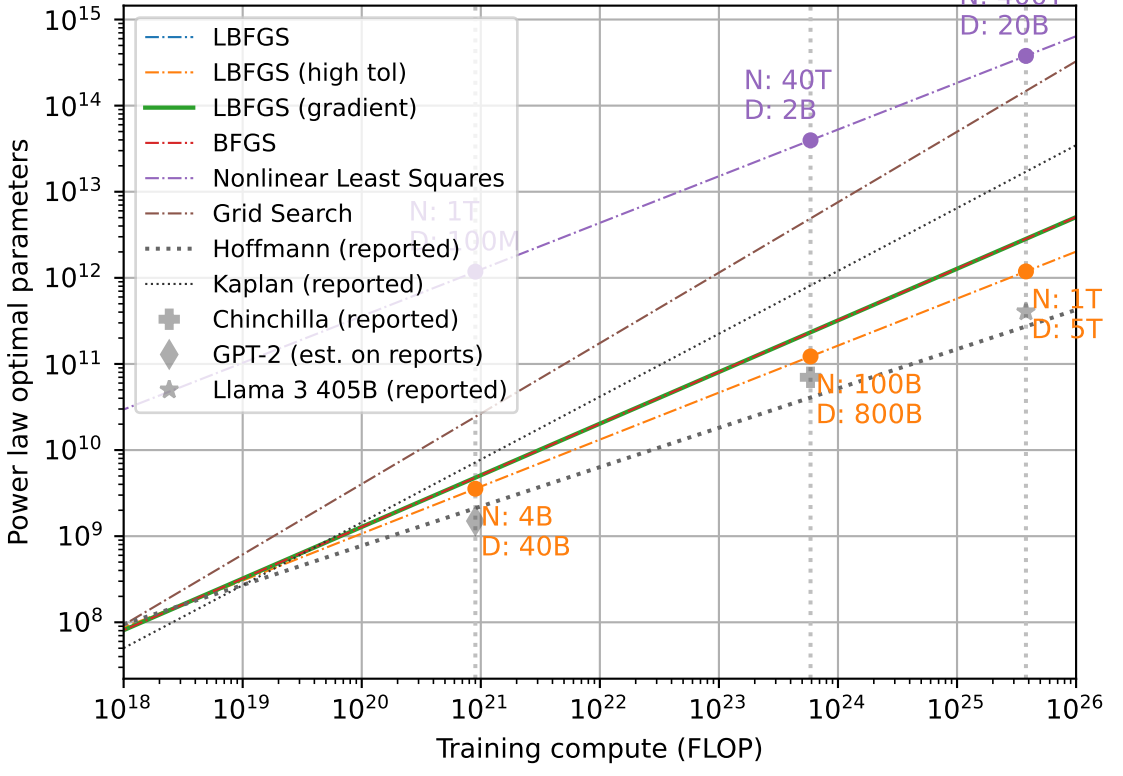
LBFGS
LBFGS (high tol)
LBFGS (gradient)
BFGS
Nonlinear Least Squares
Grid Search
Hoffmann (reported)
Kaplan (reported)
Chinchilla (reported)
GPT-2 (est. on reports)
Llama 3 405B (reported)
N: 400T
D: 20B
N: 40T
D: 2B
N: 1T
D: 100M
N: 1T
D: 5T
N: 100B
D: 800B
N: 4B
D: 40B
Power law optimal parameters
Training compute (FLOP)
10^{8}
10^{9}
10^{10}
10^{11}
10^{12}
10^{13}
10^{14}
10^{15}
10^{18}
10^{19}
10^{20}
10^{21}
10^{22}
10^{23}
10^{24}
10^{25}
10^{26}